

Stat 240 Midterm

February 2026

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PUBLISHED
February 25, 2026

```
download.file("https://owenward.github.io/files/nhl_data.tsv", "nhl_data.tsv")
read.csv("nhl_data.tsv", sep = "\t")
```

	Team	GP	SOG	W	L	OTL	PTS	Reg_PTS	PPG	SO_PTS	Pct_SO	RW	ROW	SOW	SOL
1	LA	82	9	47	25	10	104	98	1.3424658	6	0.6666667	37	41	6	3
2	STL	82	6	37	38	7	81	78	1.0263158	3	0.5000000	27	34	3	3
3	EDM	82	4	50	23	9	109	109	1.3974359	0	0.0000000	45	50	0	4
4	CGY	82	7	38	27	17	93	91	1.2133333	2	0.2857143	31	36	2	5
5	PIT	82	2	40	31	11	91	90	1.1250000	1	0.5000000	31	39	1	1
6	DET	82	7	35	37	10	80	77	1.0266667	3	0.4285714	28	32	3	4
7	PHI	82	3	31	38	13	75	73	0.9240506	2	0.6666667	26	29	2	1
8	WPG	82	2	46	33	3	95	94	1.1750000	1	0.5000000	36	45	1	1
9	SEA	82	4	46	28	8	100	100	1.2820513	0	0.0000000	37	46	0	4
10	CBJ	82	2	25	48	9	59	58	0.7250000	1	0.5000000	15	24	1	1
11	VAN	82	8	38	37	7	83	77	1.0405405	6	0.7500000	24	32	6	2
12	CHI	82	4	26	49	7	59	57	0.7307692	2	0.5000000	18	24	2	2
13	TOR	82	3	50	21	11	111	110	1.3924051	1	0.3333333	42	49	1	2
14	OTT	82	3	39	35	8	86	84	1.0632911	2	0.6666667	31	37	2	1
15	CAR	82	7	52	21	9	113	109	1.4533333	4	0.5714286	39	48	4	3
16	COL	82	9	51	24	7	109	103	1.4109589	6	0.6666667	36	45	6	3
17	FLA	82	3	42	32	8	92	90	1.1392405	2	0.6666667	36	40	2	1
18	SJ	82	7	22	44	16	60	59	0.7866667	1	0.1428571	16	21	1	6
19	MIN	82	13	46	25	11	103	96	1.3913043	7	0.5384615	34	39	7	6
20	DAL	82	7	47	21	14	108	104	1.3866667	4	0.5714286	39	43	4	3
	GF	GA	DIFF	CONF											
1	280	257	23	Western											
2	263	301	-38	Eastern											
3	325	260	65	Western											
4	260	252	8	Western											
5	262	264	-2	Eastern											
6	240	279	-39	Western											
7	222	277	-55	Eastern											
8	247	225	22	Western											
9	289	256	33	Western											
10	214	330	-116	Eastern											
11	276	298	-22	Western											
12	204	301	-97	Western											
13	279	222	57	Eastern											
14	261	271	-10	Eastern											
15	266	213	53	Eastern											
16	280	226	54	Western											
17	290	273	17	Eastern											

18 234 321 -87 Eastern
19 246 225 21 Western
20 285 218 67 Western

```
library(RSQLite)
library(DBI)
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr 1.1.4 ✓ readr 2.1.6
✓ forcats 1.0.1 ✓ stringr 1.6.0
✓ ggplot2 4.0.1 ✓ tibble 3.3.1
✓ lubridate 1.9.4 ✓ tidyr 1.3.2
✓ purrr 1.2.1

— Conflicts — tidyverse_conflicts() —

✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag() masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

Problem 1

#a

```
## write code here
data <- read.csv("nhl_data.tsv", sep = "\t")
nhl_sub <- data |> filter(W < 50)
nhl_sub
```

	Team	GP	SOG	W	L	OTL	PTS	Reg_PTS	PPG	SO_PTS	Pct_SO	RW	ROW	SOW	SOL
1	LA	82	9	47	25	10	104	98	1.3424658	6	0.6666667	37	41	6	3
2	STL	82	6	37	38	7	81	78	1.0263158	3	0.5000000	27	34	3	3
3	CGY	82	7	38	27	17	93	91	1.2133333	2	0.2857143	31	36	2	5
4	PIT	82	2	40	31	11	91	90	1.1250000	1	0.5000000	31	39	1	1
5	DET	82	7	35	37	10	80	77	1.0266667	3	0.4285714	28	32	3	4
6	PHI	82	3	31	38	13	75	73	0.9240506	2	0.6666667	26	29	2	1
7	WPG	82	2	46	33	3	95	94	1.1750000	1	0.5000000	36	45	1	1
8	SEA	82	4	46	28	8	100	100	1.2820513	0	0.0000000	37	46	0	4
9	CBJ	82	2	25	48	9	59	58	0.7250000	1	0.5000000	15	24	1	1
10	VAN	82	8	38	37	7	83	77	1.0405405	6	0.7500000	24	32	6	2
11	CHI	82	4	26	49	7	59	57	0.7307692	2	0.5000000	18	24	2	2
12	OTT	82	3	39	35	8	86	84	1.0632911	2	0.6666667	31	37	2	1
13	FLA	82	3	42	32	8	92	90	1.1392405	2	0.6666667	36	40	2	1
14	SJ	82	7	22	44	16	60	59	0.7866667	1	0.1428571	16	21	1	6
15	MIN	82	13	46	25	11	103	96	1.3913043	7	0.5384615	34	39	7	6
16	DAL	82	7	47	21	14	108	104	1.3866667	4	0.5714286	39	43	4	3
	GF	GA	DIFF				CONF								
1	280	257		23			Western								
2	263	301		-38			Eastern								

3	260	252	8	Western
4	262	264	-2	Eastern
5	240	279	-39	Western
6	222	277	-55	Eastern
7	247	225	22	Western
8	289	256	33	Western
9	214	330	-116	Eastern
10	276	298	-22	Western
11	204	301	-97	Western
12	261	271	-10	Eastern
13	290	273	17	Eastern
14	234	321	-87	Eastern
15	246	225	21	Western
16	285	218	67	Western

```
print(dim(nhl_sub)[1])
```

[1] 16

Problem 1

#b

```
nhl_mod <- select(nhl_sub, -GF, -GA, -DIFF)
nhl_mod
```

	Team	GP	SOG	W	L	OTL	PTS	Reg_PTS	PPG	SO_PTS	Pct_SO	RW	ROW	SOW	SOL
1	LA	82	9	47	25	10	104	98	1.3424658	6	0.6666667	37	41	6	3
2	STL	82	6	37	38	7	81	78	1.0263158	3	0.5000000	27	34	3	3
3	CGY	82	7	38	27	17	93	91	1.2133333	2	0.2857143	31	36	2	5
4	PIT	82	2	40	31	11	91	90	1.1250000	1	0.5000000	31	39	1	1
5	DET	82	7	35	37	10	80	77	1.0266667	3	0.4285714	28	32	3	4
6	PHI	82	3	31	38	13	75	73	0.9240506	2	0.6666667	26	29	2	1
7	WPG	82	2	46	33	3	95	94	1.1750000	1	0.5000000	36	45	1	1
8	SEA	82	4	46	28	8	100	100	1.2820513	0	0.0000000	37	46	0	4
9	CBJ	82	2	25	48	9	59	58	0.7250000	1	0.5000000	15	24	1	1
10	VAN	82	8	38	37	7	83	77	1.0405405	6	0.7500000	24	32	6	2
11	CHI	82	4	26	49	7	59	57	0.7307692	2	0.5000000	18	24	2	2
12	OTT	82	3	39	35	8	86	84	1.0632911	2	0.6666667	31	37	2	1
13	FLA	82	3	42	32	8	92	90	1.1392405	2	0.6666667	36	40	2	1
14	SJ	82	7	22	44	16	60	59	0.7866667	1	0.1428571	16	21	1	6
15	MIN	82	13	46	25	11	103	96	1.3913043	7	0.5384615	34	39	7	6
16	DAL	82	7	47	21	14	108	104	1.3866667	4	0.5714286	39	43	4	3
CONF															
1	Western														
2	Eastern														
3	Western														
4	Eastern														

5 Western
6 Eastern
7 Western
8 Western
9 Eastern
10 Western
11 Western
12 Eastern
13 Eastern
14 Eastern
15 Western
16 Western

Problem 1

#c

```
write.table(nhl_mod, file = "nhl.psv", sep="._.")
```

Problem 2

```
## write code here
```

Problem 3 (Delete if not used)

```
## write code here
```

Problem 4 (Delete if not used)

```
## write code here
```

Problem 5 (Delete if not used)

```
## write code here
```